

Electron impact ionization and transport in nitrogen–argon mixtures

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Received 4 October 2008, in final form 6 October 2008

Published 15 January 2009

Online at stacks.iop.org/JPhysD/42/045202

Abstract

The electron drift velocities and ionization coefficients in the nitrogen–argon mixtures have been measured and calculated over a wide range of E/N and mixture concentrations. It was found that standard cross section sets for low and moderate energies fit the transport data very well. Derived quantities such as the mean energy for electrons and the ratio between the transverse diffusion coefficients and the electron mobility have also been calculated.

1. Introduction

Interest in the study of kinetic phenomena in low pressure, low current gas discharges has increased recently since it has been shown that such phenomena affect the models of non-equilibrium, collisional plasmas and also because it is possible to represent them well with kinetic plasma models [1, 2]. One of the test models for a number of kinetic phenomena is the mixture of argon and nitrogen, where large vibrational excitation cross sections coincide with the low, albeit rapidly increasing elastic cross sections of argon in the region of the Ramsauer–Townsend minimum. On the other hand, argon and nitrogen were the most studied targets in swarm experiments, thereby being the prototypical atomic and molecular gases, and therefore one may expect their cross sections to be well determined [3, 4].

It used to be typical in swarm physics that studies of the cross section data were performed at low energies, where all of the available transport coefficients could be measured. The task of completing measurements at higher E/N was seldom attempted. This paper is one such attempt aimed at improving the basis for the models by measuring the electron transport coefficients, and simultaneously calculating them with the available cross section data sets. The idea is to measure drift velocities at somewhat higher E/N and also to measure ionization coefficients. The ionization coefficients may be used very efficiently to establish the total energy losses

in the region of electronic excitation/dissociation since cross sections for ionization are known quite accurately, and since the tail of the distribution function is affected very much by overall energy losses below the threshold of ionization, then the ionization rate may be used to test the overall inelastic processes at higher energies [5].

We have set out to perform a comparison between experimental data for transport coefficients at low currents and low pressures. The range of energies that was covered is broad and will enable us to venture into the mean energy range which yields significant electronic excitation. Thus, we report measured and calculated values of the electron drift velocity, the longitudinal diffusion, the ionization coefficient and the calculated values of the ratio between the transverse diffusion coefficient and the electron mobility, and the mean electron energies in the swarm in the Ar–N₂ mixtures. The density-normalized electric field, E/N , was varied over a wide range from 0.18 to 330 Td (1 Td = 10^{-17} V cm²).

It is also important to note that the gas mixture technique has often been used to reduce the problem of non-uniqueness of the cross sections obtained from the transport data, and the data for argon–nitrogen mixtures have been critical in obtaining high accuracy vibrational excitation cross sections for nitrogen. In addition, the mixture has been the primary example of negative differential conductivity (NDC) [6–9] and also of enhanced mobility caused by inelastic processes [9]. Finally, it

is important to note that calculations in such a mixture by a standard two-term (TT) technique may be inaccurate and that often a multi-term or a Monte Carlo (MC) simulation is required, but typically the accuracy of the TT results for the drift velocities is greater than the accuracy for other coefficients.

While mixtures of nitrogen and argon have not been known for their direct application, the recent use of low- k dielectrics [10, 11] may lead to the need for arranging discharges in nitrogen containing mixtures (with hydrogen) and then one could face the need to use argon as the additional buffer gas in order to optimize the electron distribution function and the collision rates.

2. Data and procedure for calculations

We have used the available cross section sets consistent with those derived by Haddad [12, 13] for low electron energies and those of Phelps at higher mean electron energies [14, 15]. The details of the cross section sets and a comprehensive list of references have been given in some of our previous papers [16, 17]. The calculations were performed both by a TT code (TT, ELENDIF) [18] and by our MC code that has been verified against all benchmark tests and was shown to give results of the highest available accuracy [19, 20]. The TT results were mainly taken as a guide and to provide smoother graphs, especially for the drift velocity, where the TT theory works well [13, 21] but all the final conclusions were based on the MC calculations.

The MC code also allowed us to generate a broader set of data including both bulk and flux versions of transport coefficients [22, 23]. The code is based on numerical integration of collision probability. The sampling of transport coefficients was performed for time of flight experiments but it is possible to sample at the same time data for different physical situations.

In the MC simulations we have followed from 10 000 to 100 000 electrons through a large number of collisions. Typically, a statistical uncertainty of better than 0.1% was achieved for most (velocity space) parameters. We may therefore conclude that the uncertainty in the calculated data comes mainly from the uncertainty of the cross sections that were employed.

3. Experiment

The swarm parameters in the Ar–N₂ mixtures were measured with a pulsed Townsend apparatus, the details of which have been given elsewhere [24]. Briefly, the pulsed Townsend method relies on the observation of the total displacement current produced by the motion of ions and electrons across a parallel-plate discharge gap that produces a highly homogeneous electric field. The initial electrons are released by the incidence of a UV pulse from a Nd–YAG laser ($\lambda = 355$ nm, 3 ns FWHM) upon an aluminium cathode. Since the mobility of the electrons is much higher than that of the ions, it is possible to separate the electronic contribution from the ionic one. In this case, the electron drift velocity, the

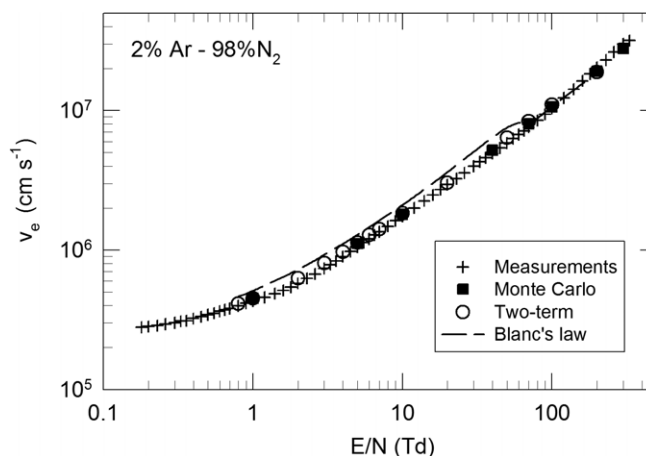


Figure 1. The drift velocity of electrons in the 2% Ar–98% N₂ mixture. Crosses, measurement; solid squares, MC calculations; open circles, TT calculations; broken line, Blanc's law (CME).

effective ionization coefficient and the density-normalized diffusion coefficient were derived from an analysis of the electron component, the details of which can be found in [15].

The measurements were carried out at room temperatures in the range 293–300 K. The gas pressure was varied between 0.5 and 600 Torr. The Ar and N₂ gas samples had a quoted purity of 99.99% and 99.999%, respectively. The Ar–N₂ mixture compositions were varied from 1% to 99.95% Ar in N₂. In this paper we discuss only the mixtures containing 2%, 50%, 80% in connection with the calculations, and those containing 99–99.95% Ar in N₂ for comparison with the measurements of Haddad [12].

The experimental uncertainty of the measured data are ± 1 –2% for the drift velocities, ± 10 % for the longitudinal diffusion coefficients ± 4 –7% for the ionization coefficients.

4. Results

4.1. Electron drift velocities

The electron drift velocities in Ar–N₂ mixtures containing 2%, 50% and 80% Ar are shown in figures 1–3, respectively. As abundances move towards smaller percentages of the molecular gas, the effect of inelastic collisions on the electron drift velocity becomes more and more pronounced, forming an elbow in the E/N dependence. Still at 20% of N₂ content, NDC does not develop. In these figures we compare the experimental data with the results of TT and MC calculations. We also give a prediction of the mixture results from the data for pure gases by using a common mean energy (CME) corrected Blanc's law [25, 26].

We see from these plots that both the TT (ELENDIF) and the MC calculations agree well with each other and with the measurements over the whole range of E/N . This is in agreement with the results of Haddad [12], who found that multi-term correction for the drift velocity in N₂–Ar mixtures may be up to 1.5%, which is of the order of the uncertainty of our experimental data. On the other hand, the application of Blanc's law to this mixture from the values of the electron drift

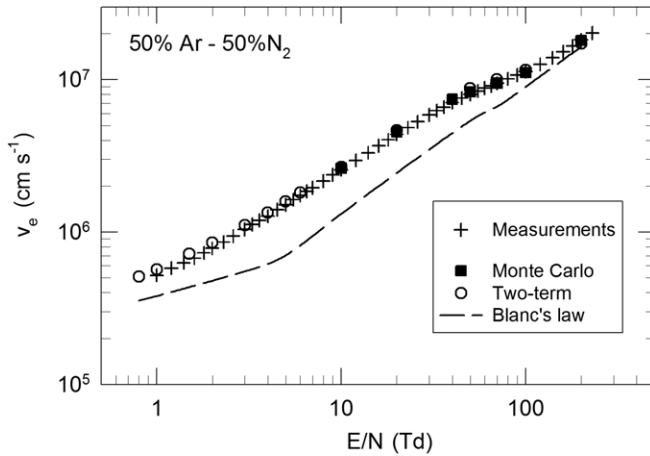


Figure 2. The drift velocity of electrons in the 50% Ar–50% N₂ mixture. Crosses, measurement; solid squares, MC calculations; open circles, TT calculations; broken line, Blanc's law (CME).

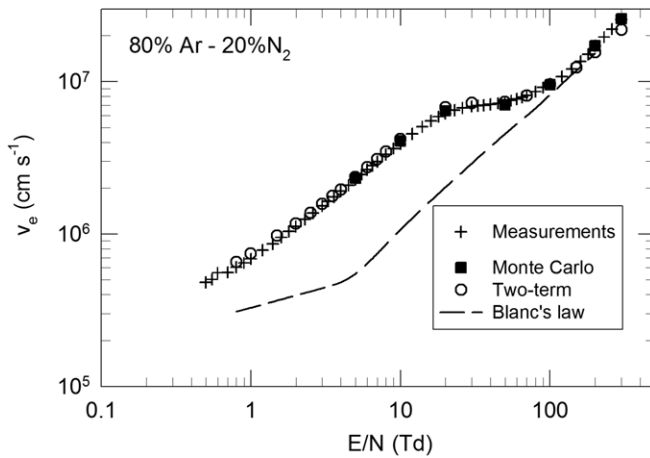


Figure 3. The drift velocity of electrons in the 80% Ar–20% N₂ mixture. Crosses, measurement; solid squares, MC calculations; open circles, TT calculations; broken line, Blanc's law (CME).

velocity in pure Ar and N₂ fails to reproduce the measured and calculated values previously referred to.

Blanc's law fails more seriously as the abundance of N₂ decreases. This combination of gases is one of the most difficult cases for application of Blanc's law, since the inelastic collisions in Ar start at high threshold (around 11 eV), while vibrational and rotational excitation in N₂ control electron energy at much lower energies (around 0.02 eV). Mean energies in argon at the same E/N are much higher than those in pure nitrogen, even by an order of magnitude [26]. At the same time as the abundance of molecular gas in argon is smaller, the effect of molecular gas as compared with the pure atomic gas is greater. This leads to a significant difference of energy distribution functions, making the application of Blanc's law impossible [27, 28]. Application of the CME corrected Blanc's law, where one combines transport data at the same mean energy and not at the same E/N , improves the agreement, but then the different shapes of the distribution functions may affect the applicability of Blanc's law.

Figure 4 shows the present measurements of the electron drift velocity for 99%–99.95% Ar in N₂, respectively, together

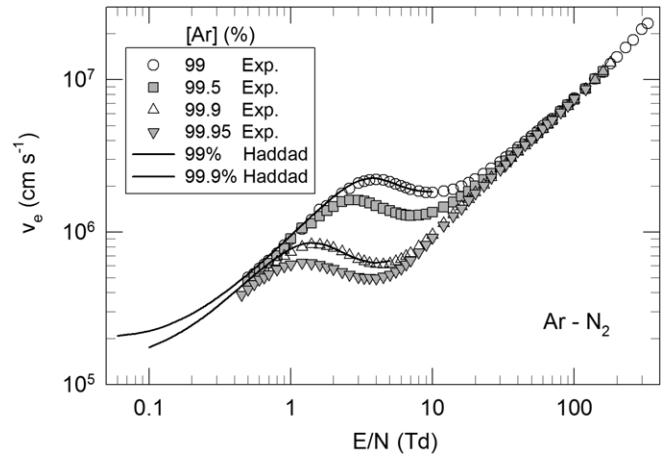


Figure 4. The drift velocity of electrons in Ar–N₂ mixtures for small N₂ content. Figures in the inset correspond to Ar concentrations in the mixture. Symbols correspond to measurements, while solid lines correspond to the trend of the measurements of Haddad [12].

with previous measurements of Haddad [13] for 99% and 99.9% Ar. In both cases, the agreement with the presently measured data is excellent, well below our stated experimental uncertainty. In this context, it is interesting to mention that such agreement is irrespective of the technique used. Haddad used a time of flight drift tube which has been known to give the experimental data of the highest accuracy achieved so far [12, 13], but the experiment was limited to low values of mean electron energy. The present measurements were obtained from a pulsed Townsend experiment. This agreement, far from being fortuitous, is the result of two carefully laid out measurements and it gives support to the quality of our measurements. One should bear in mind that the differences between the experimental results of different types of experiments may be expected only when non-conservative collisions become significant [29, 30] and, in this case, only when ionization becomes important. Haddad did not extend his measurements to such high fields but it would be, in principle, interesting to extend the comparison with the region where non-conservative processes are important (either to high E/N where ionization is significant or to electronegative gases).

We have made sure that the transport of electrons is studied for conditions where there are no non-hydrodynamic effects or, in other words, that the electron swarm is properly relaxed throughout most of the drift tube (and also in the simulations). In order to illustrate the sensitivity of the present experimental setup, also shown in figure 4 is the measurement of the electron drift velocity for 99.95% Ar in 0.05% N₂. The curve which lies even closer to the curve for pure Ar still bears a clear signature of a NDC effect. Such NDC effect has been discussed thoroughly before [6–8]. In connection with the present experiment with the NDC, it was discussed for electron transport in He–Xe mixtures [26].

In general, NDC is observed for all mixtures (covered by our study) with a small abundance of N₂. Mixtures of argon and nitrogen are one of the classic examples for NDC where it was first studied and at the same time advantage of NDC was taken in order to obtain accurate cross sections [13]. At this point it is worth noting that the NDC effect, which

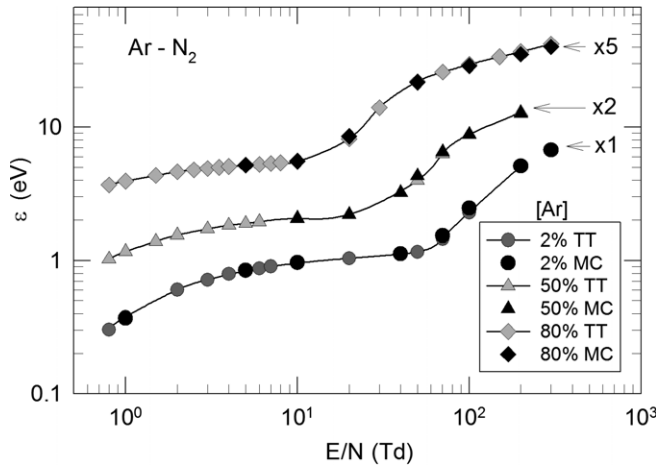


Figure 5. The calculated mean energy of electrons in N_2 -Ar mixtures for 2%, 50% and 80% Ar content by the TT and the MC methods. The multipliers shown to the right of the curves indicate the factor by which the actual values were multiplied in order to ease their appreciation on a single plot.

increases as the abundance of nitrogen diminishes, should start decreasing at exceedingly small abundances of the molecular gas. Nevertheless the same reasons leading to the failure of the Blanc's law at small N_2 abundances contribute to NDC. In any case, it is important for us that fitting such a complex structure requires good agreement of both momentum balance and energy balance controlling cross sections, i.e. of the elastic cross section and of the inelastic cross sections that dominate energy exchange in this range of E/N . The fact that a good fit is obtained by a set which is mainly an extension of that of Haddad [13] may lead some to claim that this was to be expected, but one should bear in mind that these are two completely independent sets of data obtained by two very different techniques and therefore good agreement of calculated and experimental data indicates excellent internal consistency of both the cross sections and of the experimental techniques [5, 16]. In spite of complexity and a large number of different abundances we have achieved good fits of experimental data by using single sets of the cross sections for nitrogen and argon.

4.2. Mean energy of electrons

Even though measurements of the mean electron energy in the swarm are inaccessible to the present experiments, the calculations do permit us to obtain accurate results. Figure 5 shows the variation of the mean energy of the electron swarm for the Ar- N_2 mixtures containing 2%, 50% and 80% Ar, over a wide range of E/N . Note the excellent agreement between the mean energies calculated by the TT method and those obtained from a MC calculation. While the addition of nitrogen reduces the mean energy for energies below the threshold for excitation, the effects observed in the figure 5 are not significant since in all cases the amount of nitrogen is still large. Thus a greater effect is provided by the multipliers which had to be applied to avoid possible overlap. Significant effects of the abundance of molecular gas on the mean energy may be observed for low abundances of nitrogen of the order of 1%. However, under

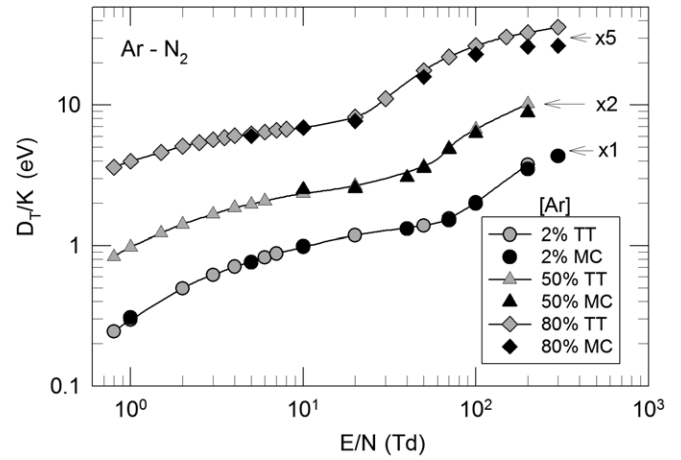


Figure 6. The calculated ratio D_T/K between the transverse diffusion coefficient and the electron mobility for electrons in N_2 -Ar mixtures for 2%, 50% and 80% Ar content by the TT and the MC methods. The multipliers shown to the right of the curves indicate the factor by which the actual values were multiplied in order to ease their appreciation on a single plot.

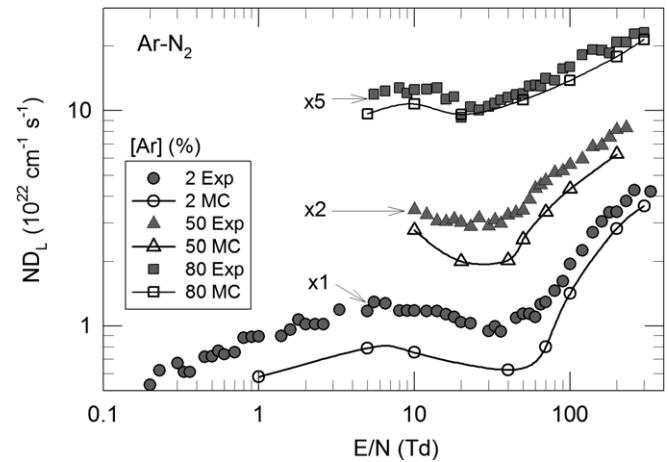


Figure 7. The density-normalized longitudinal diffusion coefficients of electrons in N_2 -Ar mixtures for 2%, 50% and 80% Ar. Closed symbols, measured values; open symbols, MC calculations. The multipliers shown at the left of the curves indicate the factor by which the actual measured and calculated values were multiplied in order to ease their appreciation on a single plot.

those circumstances the effect has been shown through the behaviour of the characteristic energy by Haddad [13]

4.3. Diffusion coefficients

Also inaccessible to the present experimental setup (but not in general) is the measurement of the transverse diffusion coefficient D_T or of its ratio with the electron mobility, D_T/K , where $K = v_e/E$. The calculated values of D_T/K are plotted in figure 6 for the same mixture compositions as above. The TT and MC calculations seem to agree with each other although here one can see discrepancies at large E/N which are consistent with the predictions of Haddad [13].

We have, however, been able to measure the longitudinal diffusion coefficient, D_L , presented here in figure 7 as the

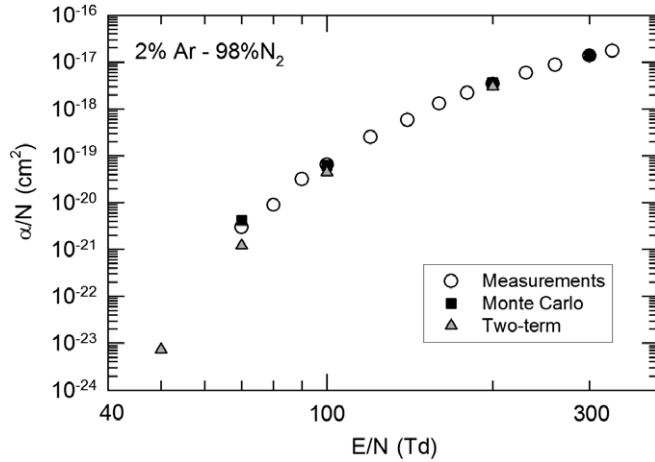


Figure 8. The ionization coefficient of electrons in the 2% Ar–98% N₂ mixture. Open circles, measurement; solid squares, MC calculations; open circles, TT calculations.

normalized product ND_L , as a function of E/N for the 2%, 50% and 80% Ar–N₂ mixtures. A comparison with the calculated values shows that in all cases the measurements follow the same trend as the calculated data, although the experimental results are systematically higher, turning out to be up to about a factor of 1.5 higher for $E/N < 100$ Td. In general, the agreement between measurements and calculations improves over the high E/N region. The discrepancy could indicate a need to improve the cross section further, but from our previous experience it has been very difficult to achieve fits of ND_L as there is also a possibility that experimental data could be affected in a systematic way. In most situations, running experiments optimized to measure drift velocities leave a greater possibility for error in measurements of diffusion coefficients and vice versa. On the experimental side, the measurement of longitudinal diffusion is affected by the convolution between the finite duration of the laser pulse (3–5 ns) and the instrumental bandwidth (40 MHz), since the most useful portion of the electron transient to determine longitudinal diffusion is its falling edge, from which the electron transit time and hence the electron drift velocity are determined accurately. Thus, it is most likely that the measured longitudinal diffusion be affected systematically by these two limiting factors. On the other hand, the falling edge of the pulse is enhanced at low pressures; however, we cannot lower the pressure indefinitely, since then one would have an appreciable influence of the electric field produced by the avalanche itself, thereby affecting the overall electric field and hence E/N . Sometimes, parameters for optimal measurements are difficult to access. Thus we have not attempted to modify the cross sections as those were based on more reliable data for drift velocities and D_T/μ for pure gases and their mixtures.

4.4. Ionization coefficients

The measured and calculated density-normalized ionization coefficients α/N for the above mentioned mixtures are given in figures 8–10. We observe very good agreement between the measurements and both the TT and MC calculations

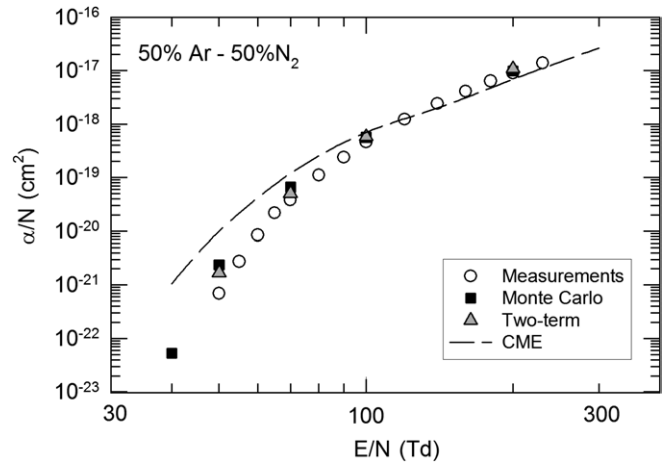


Figure 9. The ionization coefficient of electrons in the 50% Ar–50% N₂ mixture. Open circles, measurement; solid squares, MC calculations; open circles, TT calculations; broken line, Blanc's law (CME).

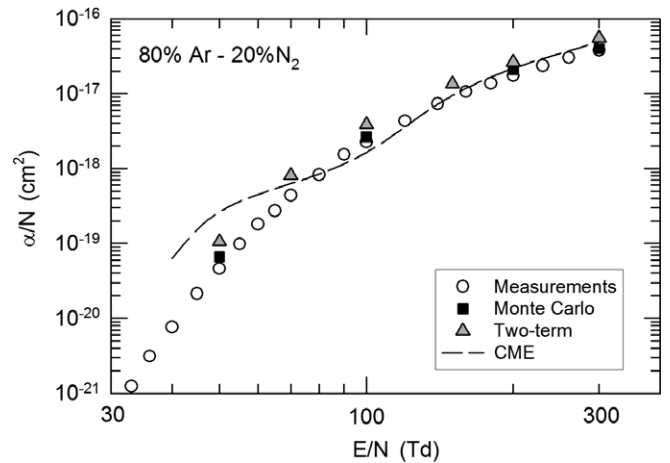


Figure 10. The ionization coefficient of electrons in the 80% Ar–20% N₂ mixture. Open circles, measurement; solid squares, MC calculations; open circles, TT calculations; broken line, Blanc's law (CME).

for $E/N > 70$ Td. For $E/N < 70$ Td there is an increasing discrepancy which is stronger for the TT result. This could be due to the failure of the TT procedure for processes with high thresholds [31] but also due to numerical problems as values of ionization at those E/N become very small. MC agrees better with experiment but still it reveals a small systematic discrepancy that could be probably fixed by adjusting to some degree the electronic excitation cross sections. However the discrepancy is small enough that it did not justify further modifications again, since the experimental uncertainty becomes larger for small ionization coefficients. Thus, there is overall agreement between MC and experimental results, which is thought to be very good and, except for the lowest E/N , the TT calculations produce reasonably good results as well.

We have also added predictions based on a mixture law for ionization rates which is analogous to Blanc's law, as well as having used a CME procedure here [27] as that employed for the calculation drift velocities [26]. As can be seen in

figures 8–10, the CME procedure fails at low E/N values. At the lower range of mean electron energies, ionization of N_2 dominates over that of Ar. On the other hand, even for the same mean electron energy, distribution functions are not quite identical in pure gases and in the mixture. The main differences are in the high energy tail which is responsible for ionization. Argon is a difficult gas to use because the lack of inelastic energy losses means that electrons gain high energies only to face large inelastic losses at very high energies. Therefore, the distribution function has a very sharp drop at the high energy tail. The data for the mixture is a combination of the results for two pure gases which therefore have two very different high energy tails, even though their mean energies are the same. Thus, at lower energies, where N_2 prevails in ionization, the ionization coefficients are overestimated due to the extended high energy tail in the electron energy distribution functions for pure N_2 as compared with what would be relevant for the mixture. When the mean energy increases and the overlap of the distribution function with the ionization (and other cross sections with high energy losses) becomes more substantial, then the agreement improves substantially as the high energy tails develop both in pure argon and of course in the mixture. By increasing the abundance of N_2 in the mixture, the electron energy distribution function for the mixture becomes more similar to that in pure N_2 , and hence the agreement of the CME procedure with the experiment and the TT and MC calculations is also better [27].

5. Conclusions

The purpose of this paper is to improve the basis for the Ar/ N_2 mixture models by measuring the electron transport coefficients over a wide range of E/N . At the same time, the transport coefficients were calculated by a TT code and a MC technique. In all cases, except for ionization coefficients, the agreement between the TT code and the MC technique is excellent.

Numerical calculations and experimental results for the drift velocities of electrons were supplemented by standard Blanc's law procedure. For electrons at moderate and even small values of E/N , Blanc's law cannot be used with any degree of reliability and it is not able to predict NDC. However, for higher values of E/N Blanc's law becomes as good as both numerical calculations since the differences between mean energies for different gases at the same E/N are small.

In the case of the longitudinal diffusion, the discrepancies between experimental data and MC results are significant. The discrepancies are even larger at low and intermediate fields and increase with the increase in N_2 abundance. It appears that, on the one hand, the experimental conditions were such that they favour accurate determination of the drift velocity while the accuracy of the diffusion coefficient is then compromised and, on the other hand, it is also possible that anisotropy of scattering needs further adjustment to fit the longitudinal diffusion as well. Nevertheless, we did not feel justified to modify the cross sections obtained on the basis of very accurate low energy data without proving the internal consistency of our longitudinal diffusion results. The transverse diffusion

coefficients are not available for our range of E/N and the calculated D_T/K results may be a useful source of data since the MC and TT calculations agree within the expected limits for the failure of the TT approximation.

Finally, the measured ionization coefficients are compared with the calculated data and reasonable agreement for all mixtures is reached in the case of the MC technique. CME Blanc's law procedure disagrees significantly at low values of E/N . It is observed that with increasing concentration of Ar atoms the discrepancies are even larger.

Acknowledgments

This work was supported by DGAPA-UNAM, Project IN 108508 and by MNTRS 141025 project. Thanks are due to A Bustos, E Ortíz and R Bustos for technical assistance.

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